

Evaluating the impact of explainable AI on clinicians' decision-making: A study on ICU length of stay prediction

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ABSTRACT

Background: Explainable Artificial Intelligence (XAI) is increasingly vital in healthcare, where clinicians need to understand and trust AI-generated recommendations. However, the impact of AI model explanations on clinical decision-making remains insufficiently explored.

Objectives: To evaluate how AI model explanations influence clinicians' mental models, trust, and satisfaction regarding machine learning-based predictions of Intensive Care Unit (ICU) Length of Stay (LOS).

Methods: This retrospective mixed-methods study analyzed electronic health record data from 8,579 patients admitted to a surgical ICU in South Korea between 2019 and 2022. Seven machine learning models were developed and evaluated to predict ICU LOS at 2-hour intervals during the initial 12 hours post-admission. The Random Forest (RF) model in the 10- to 12-hour window, with an AUROC of 0.903, was selected for explanation using SHapley Additive exPlanations. Fifteen ICU clinicians assessed four distinct types of explanations ('Why', 'Why not', 'How to', and 'What if') via web-based experiments, surveys, and interviews.

Results: Clinicians' feature selections aligned more closely with the RF model after explanations, as demonstrated by an increase in Spearman correlation from -0.147 ($p = 0.275$) to 0.868 ($p < 0.001$). The average trust score improved from 2.8 to 3.9. The average satisfaction scores for the 'Why', 'Why not', 'How to', and 'What if' explanations were 3.3, 3.8, 3.6, and 4.1, respectively.

Conclusion: AI model explanations notably enhanced clinicians' understanding and trust in AI-generated ICU LOS predictions, although complete alignment with their mental models was not achieved. Further refinement of AI model explanations is needed to support better clinician-AI collaboration and its integration into clinical practice.

1. Introduction

Explainable Artificial Intelligence (XAI) aims to develop AI systems that transparently communicate decision-making processes to humans, including rationale, potential limitations, and anticipated outcomes [1]. The demand for XAI is particularly strong in healthcare, where high-stakes decisions directly affect patient safety and outcomes [2]. In this context, clinicians and other stakeholders require clear explanations to

understand AI-generated recommendations [3,4]. Lack of transparency in AI models often creates hesitancy among clinicians, highlighting the importance of explainability for building trust and effective integration into clinical practice [5]. A previous study reported that clinicians' acceptance of AI-generated recommendations correlates with their understanding of the underlying reasoning provided by the AI models [6]. Therefore, enhancing AI explainability is critical to fostering stakeholders' trust, facilitating successful AI adoption in clinical settings, and

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ultimately improving patient outcomes [7,8].

Considering public policy and regulatory frameworks, global recognition of the importance of XAI is increasing, particularly in healthcare, where decisions have significant consequences. The European Union (EU)'s General Data Protection Regulation (GDPR) emphasizes individuals' rights to explanations for AI-driven decisions using personal data, underscoring the importance of transparency and accountability in AI systems [9]. Moreover, the EU's AI Act mandates that high-risk AI systems provide clear, understandable, and traceable information [10]. The World Health Organization (WHO) similarly advocates that healthcare-related AI systems should be explainable to stakeholders, highlighting the need to balance explainability with accuracy, as excessive complexity can create confusion and frustration for both explainers and users [11].

Despite notable progress in healthcare-focused XAI research, the practical application of XAI in clinical settings remains challenging due to ongoing debates about the actual usefulness of explainability in real-world AI implementations [12]. A fundamental issue is that explanation efficacy — how well an explanation functions — depends not solely on the model but largely on how users perceive and accept it [13]. Simply providing full model transparency does not guarantee user comprehension and may overwhelm clinicians [12]. Effective explanations require presenting relevant information clearly and concisely, tailored to users' ability to gain new insights, the explanation's intended purpose, and various human factors [14].

The Defense Advanced Research Projects Agency (DARPA) initiated the XAI program to develop Machine Learning (ML)-based AI systems that are understandable and evaluable by human users [1]. The program emphasizes evaluating explanation effectiveness through five key measurement concepts: mental models, user satisfaction, trust assessment, task performance, and correctability [15]. A prior systematic review based on DARPA's criteria analyzed healthcare studies employing XAI techniques but identified notable shortcomings, such as the lack of standardized methods and inconsistent measurement tools [16]. Specifically, no studies assessed mental models — critical for users to comprehend, anticipate, and mentally simulate how an AI system operates, including understanding its limitations [17,18].

This study aimed to explore the impact of AI model explanations on clinicians' view of AI-generated decisions (i.e., "mental models"), trust in the AI model, and satisfaction with the explanations following DARPA's XAI evaluation framework [19]. This study assessed how XAI methods affect clinicians' acceptance of AI-generated results with AI model explanations using Length of Stay (LOS) prediction in the surgical Intensive Care Unit (ICU) as a use case.

2. Materials and methods

2.1. Development of the ICU LOS prediction model

2.1.1. Study population

This retrospective study analyzed electronic health record (EHR) data from 10,000 consecutive patients admitted to the surgical ICU of a 1,761-bed tertiary teaching hospital in South Korea between January 1, 2019 and December 31, 2022. Inclusion criteria were adult patients admitted to the surgical ICU and subsequently transferred to general wards within the same hospital. After excluding 1,421 patients due to death in the surgical ICU, transfer to other institutions without clinical improvement, or transfer to general wards within 12 hours of admission, 8,579 patients remained for analysis.

2.1.2. Data pre-processing

The dataset was restructured into 2-hour intervals from ICU

admission to 12 hours post-admission to develop an optimal predictive model for identifying critical time windows. Variables with more than 50 % missing values were excluded. Missing data (< 50 %) within the first 2 hours were handled through iterative imputation, followed by forward imputation. Categorical input variables were transformed into binary variables using one-hot encoding. The continuous output variable was discretized into two groups: short ICU LOS (≤ 60 hours, class 0) and prolonged ICU LOS (> 60 hours, class 1). The threshold of 60 hours was selected based on comparative evaluations across multiple thresholds, demonstrating balanced performance with high Area Under the Receiver Operating Characteristic curve (AUROC) and clinically meaningful Area Under the Precision-Recall Curve (AUPRC) — metrics particularly relevant to clinical decision-making — while also corresponding to the average surgical ICU LOS in the study cohort (Supplementary Fig. 1).

To address class imbalance, the Synthetic Minority Oversampling Technique (SMOTE) was applied to generate minority class data. Recursive Feature Elimination (RFE) was employed to identify the most relevant variables from the original 86 variables. The selected variables were then reviewed by clinical experts and normalized using min-max normalization.

2.1.3. Model training and evaluation

The dataset was randomly split into training (80 %) and testing (20 %) sets. Seven Machine Learning (ML) algorithms — Artificial Neural Network (ANN), Decision Tree (DT), K-Nearest Neighbors (KNN), Logistic Regression (LR), Random Forest (RF), Support Vector Machine (SVM), and eXtreme Gradient Boosting (XGB) — were trained using the training set. Hyperparameters were optimized using randomized search with stratified 5-fold cross-validation. Model performance was evaluated on a held-out test set using accuracy, sensitivity, specificity, AUROC, and AUPRC to select the best-performing model for explanation analysis. All models were built and analyzed using Python (Version 3.12.4) [20].

2.1.4. Model interpretability

SHapley Additive exPlanations (SHAP), based on Shapley values from game theory, were applied to quantify each feature's contribution to the prediction outcome and its influence within the model [21]. Shapley values were then used to select and visualize the top 20 variables influencing ICU LOS, following the intelligibility queries of 'Why', 'Why not', 'How to', and 'What if' [22,23].

2.2. Evaluation of explanation effectiveness

2.2.1. Participants and procedure

Fifteen ICU clinicians (minimum 1 year experience) were recruited via snowball sampling. A mixed-methods approach included web-based experiments, surveys, and interviews.

2.2.2. Measurement of clinicians' mental models

Clinicians initially predicted whether ICU LOS would exceed 60 hours based on patient information provided in a clinical scenario. Clinicians then ranked the importance of 20 features through closed card sorting using OptimalSort, constructing initial clinicians' mental models [24]. After receiving the ICU LOS prediction explanations, clinicians reviewed the same scenario, updated predictions, and revised the clinicians' mental models accordingly [25,26].

2.2.3. Measurement of clinicians' trust

Clinicians' trust in the ICU LOS prediction model was measured before and after explanations, using the DARPA-proposed 5-point trust

scale for the XAI context [15,27].

2.2.4. Explanations of the ICU LOS prediction model

Explanations of the ICU LOS prediction model using the SHAP technique were categorized into four types: (1) ‘Why’: clinicians reviewed the mean absolute SHAP plot illustrating each feature’s average contribution to predictions [28]; (2) ‘Why not’: clinicians examined the SHAP summary plot to understand how feature values influenced outcomes. This plot visually represents feature impact, with colors indicating the magnitudes of feature values and positions showing positive or negative contributions [28]; (3) ‘How to’: clinicians compared predictions by examining similar or contrasting patient cases; and (4) ‘What if’: clinicians used an interactive web interface to modify input values and observe changes in predictions, assessing the likelihood of ICU LOS exceeding 60 hours through updated risk scores [29] (Supplementary Fig. 2). The interactive interface, resembling spreadsheets with heat maps of Shapley values, was adapted from previous studies [29,30] and developed using R Shiny (Version 1.8.1) [31]. Additionally, the interface provides all 57 input variables to help clinicians gain a comprehensive understanding.

2.2.5. Measurement of clinicians’ satisfaction

Clinicians’ satisfaction with the four types of explanations was measured using the DARPA-proposed 5-point satisfaction scale for model explanations [15,27].

2.2.6. Participant interviews

Semi-structured interviews were conducted to collect clinicians’ feedback regarding the persuasiveness and potential clinical applicability of the ICU LOS prediction model explanations.

3. Results

3.1. Participant characteristics

General patient characteristics were analyzed using independent sample t-tests for continuous variables and proportion tests for categorical variables (Table 1). Among the 8,579 patients, 1,936 (22.6 %) had an ICU LOS exceeding 60 hours. Patients with prolonged ICU LOS had significantly higher HR, BE, bilirubin, BUN, chloride, creatinine, AST, ALT, aPTT, PT, and INR, as well as lower platelet counts, ESR, pain scores, and amount of O2 supply. Additionally, prolonged ICU LOS was significantly associated with abnormal ECG rhythms, restraints, positive CAM-ICU results, pressure injuries, endotracheal/tracheal tubes, gastric tubes, surgical drains, and central catheters. Patients were also significantly more likely to receive RBC, FFP, and PLT transfusions, as well as CRRT and ECMO treatments. All of these differences were statistically significant.

3.2. Development of the ICU LOS prediction model

A total of 57 selected features were used to develop the ICU LOS prediction models. Seven ML algorithms were compared at 2-hour intervals within the first 12 hours after ICU admission (Supplementary Table 1 and Fig. 3). The RF model demonstrated the best performance across most time windows, with AUROC values ranging from 0.878 to 0.903. Notably, the RF model achieved the highest AUROC of 0.903 at 10–12 hours after ICU admission, followed by the XGB (0.902), ANN (0.864), and LR (0.861) models (Fig. 1).

3.3. Evaluation of explanation effectiveness

Fifteen clinicians (13 nurses and 2 physicians, averaging 8 years of ICU experience) evaluated the effectiveness of RF model explanations. Clinicians ranked the importance of 20 selected features from 57 patient information variables presented in the scenario, both before and after

Table 1
Patient characteristics.

Variables	Total (N = 8579)	ICU LOS		p-value
		≤60 hours (N = 6643)	>60 hours (N = 1936)	
Age (year)		61.7 ± 16.4	63.8 ± 15.6	<.001
Sex				
Male	4891	3697(55.7)	1194(61.7)	<.001
Female	3688	2946(44.3)	742(38.3)	
Admission Route				
Transfer from General Wards	7722	6148(92.5)	1574(81.3)	<.001
Emergency Room	857	495(7.5)	362(18.7)	
Body Mass Index (BMI) (kg/m²)		26.0 ± 17.5	25.8 ± 18.8	0.669
Operation Duration (minutes)		184 ± 146	188 ± 184	0.244
Vital Signs				
Heart Rate (HR) (beats/minute)		72.9 ± 25.4	75.2 ± 28.7	<.001
Systolic Blood Pressure (SBP) (mmHg)		102 ± 30.5	101.3 ± 31.5	0.135
Diastolic Blood Pressure (DBP) (mmHg)		63.5 ± 11.8	62.7 ± 12.3	0.010
Mean Blood Pressure (MBP) (mmHg)		70.6 ± 17.7	70.3 ± 18.8	0.565
Body Temperature (BT) (°C)		36.8 ± 0.7	36.7 ± 0.8	<.001
SpO2 (%)		98.1 ± 0.9	98.1 ± 1.2	0.733
Laboratory Tests				
pH		7.40 ± 0.05	7.39 ± 0.08	<.001
pCO2 (mmHg)		38.2 ± 12.1	36.6 ± 13.5	<.001
pO2 (mmHg)		159 ± 105	163 ± 110	0.137
HCO3 (mmol/L)		22.7 ± 3.6	21.1 ± 4.9	<.001
Base Excess (BE) (mmol/L)		2.6 ± 2.0	4.1 ± 2.9	<.001
SaO2 (%)		98.4 ± 1.3	98.5 ± 1.5	<.001
White Blood Cell (WBC) (×10 ³ /μL)		10.7 ± 5.6	11.8 ± 6.8	<.001
Hemoglobin (g/dL)		9.6 ± 4.0	9.4 ± 4.6	0.013
Hematocrit (%)		32.6 ± 6.8	30.1 ± 9.1	<.001
Platelet (×10 ³ /μL)		172 ± 97	157 ± 125	<.001
Absolute Neutrophil Count (ANC) (×10 ³ /μL)		9377 ± 7098	9358 ± 7254	0.917
Albumin (g/dL)		3.3 ± 0.6	3.2 ± 0.8	<.001
Bilirubin (mg/dL)		0.9 ± 0.7	2.0 ± 2.3	<.001
Blood Urea Nitrogen (BUN) (mg/dL)		14.0 ± 9.1	18.9 ± 12.0	<.001
Calcium (mg/dL)		8.2 ± 0.8	8.4 ± 1.1	0.001
Chloride (mmol/L)		46.2 ± 44.0	51.0 ± 46.4	<.001
Creatinine (mg/dL)		1.0 ± 1.1	1.1 ± 1.1	0.545
estimated Glomerular Filtration Rate (eGFR) (mL/min/1.73 m ²)		89.9 ± 65.6	79.0 ± 72.4	0.078
Aspartate Aminotransferase (AST) (U/L)		29.6 ± 61.6	114 ± 246	<.001
Alanine Aminotransferase (ALT) (U/L)		25.1 ± 59.4	92.6 ± 184	<.001
Phosphorous (mg/dL)		3.6 ± 1.1	3.7 ± 1.3	<.001
Potassium (mmol/L)		4.1 ± 0.5	4.1 ± 0.6	<.001
Sodium (mmol/L)		136 ± 17.7	134 ± 21.3	<.001
Urine Analysis (UA) (mg/dL)		4.1 ± 2.0	4.6 ± 2.3	<.001
Activated Partial Thromboplastin Time (aPTT) (seconds)		28.7 ± 9.6	32.9 ± 12.3	<.001
Prothrombin Time (PT) (seconds)		12.7 ± 2.08	14.6 ± 4.4	<.001
International Normalized Ratio (INR)		1.3 ± 0.3	1.5 ± 0.5	<.001
Pain Score		2.8 ± 2.4	1.9 ± 2.3	<.001
O2 Supply		22.5 ± 16.7	15.8 ± 19.6	<.001
Electrocardiogram (ECG)				
Normal Sinus Rhythm	6834	5932(82.6)	902(46.6)	<.001
Abnormal Rhythm	1745	1251(71.4)	494(25.4)	
Restraint				
Yes	752	344(5.2)	408(21.2)	<.001
No	7827	6309(94.8)	1288(78.8)	
Confusion Assessment Method for the Intensive Care Unit (CAM-ICU)				
Positive	46	33(0.5)	13(0.7)	<.001
Negative	6066	4778(71.9)	1288(66.5)	

(continued on next page)

Table 1 (continued)

Variables	Total (N = 8579)	ICU LOS		p-value
		≤60 hours (N = 6643)	>60 hours (N = 1936)	
Not Applicable	2467	1832(27.6)	635(32.8)	
Pressure Injury				
Yes	151	90(1.4)	61(3.2)	<.001
No	8428	6554(98.6)	1875(96.8)	
Endotracheal/Tracheal Tube				
Yes	1163	457(6.9)	706(36.5)	<.001
No	7416	6186(93.1)	1230(63.5)	
Gastric Tube				
Yes	2732	1604(24.1)	1128(58.3)	<.001
No	5847	5039(75.9)	808(41.7)	
Surgical Drain				
Yes	1633	1237(18.6)	396(20.5)	0.076
No	6946	5406(81.4)	1540(79.5)	
Chest Tube				
Yes	926	808(12.2)	118(6.1)	<.001
No	7653	5835(87.8)	1818(93.9)	
Foley Catheter				
Yes	7957	6221(93.6)	1736(89.7)	<.001
No	622	422(6.4)	200(10.3)	
Central Catheter				
Yes	5125	3718(56.0)	1407(72.7)	<.001
No	3454	2925(44.0)	529(27.3)	
Arterial Catheter				
Yes	6994	5447(82.0)	1547(79.9)	0.040
No	1585	1196(18.0)	389(20.1)	
Red Blood Cell (RBC) Transfusion				
Yes	780	569(8.6)	211(10.9)	0.002
No	7799	6074(91.4)	1725(89.1)	
Fresh Frozen Plasma (FFP) Transfusion				
Yes	366	200(3.0)	166(8.6)	<.001
No	8213	6443(97.0)	1770(91.4)	
Platelet (PLT) Transfusion				
Yes	362	179(2.7)	183(9.5)	<.001
No	8217	6464(97.3)	1753(90.5)	
Continuous Renal Replacement Therapy (CRRT)				
Yes	37	2(0)	35(1.8)	<.001
No	8542	6641 (100.0)	1901(98.2)	
Extracorporeal Membrane Oxygenation (ECMO)				
Yes	1	1(0)	0(0)	1.00
No	8578	6642 (100.0)	1936 (100.0)	
Treated Nitro Oxide (NO)				
Yes	8	0(0)	8(0.4)	<.001
No	8571	6463 (100.0)	1928(99.6)	

Values are presented as number (%), mean ± standard deviation.
ICU: Intensive Care Unit, LOS: Length of Stay.

explanations. The frequency of each feature's selection was divided by its average rank. These values, along with the mean absolute Shapley values derived from the RF model, were normalized between 0 and 1 and visualized using a radar chart (Fig. 2). Initially, clinicians identified different important features compared to the RF model, but after viewing the explanations, their selections became more closely aligned.

Spearman rank correlation between clinicians' perceived feature importance and RF model-derived Shapley values improved significantly from weak and negative ($\rho = -0.147$, $p = 0.275$) to strong and positive ($\rho = 0.868$, $p < 0.001$) following explanations. Ranking difference between clinicians and the RF model was normalized on a scale from 0 to 1 before and after explanations (Supplementary Method 1). The average ranking difference decreased notably from 0.740 to 0.269 after explanations, reflecting increased alignment. A scatter plot (Fig. 3) illustrates this impact, indicating clinicians' ICU experience levels along with the reduced ranking difference after explanations.

Fig. 4 summarizes clinicians' trust evaluations of the RF model. Average trust scores increased from 2.8 to 3.9 after explanations. The greatest improvement occurred in perceived predictability of model output (average score rising from 2.3 to 4.1, an increase of 1.87 points). Conversely, concerns regarding model use showed the least change, with a decrease of 0.8 points (from 3.9 to 3.1). Additionally, clinicians assessed four explanation types generated via SHAP (Fig. 5), with mean satisfaction scores as follows: 'Why' (3.3), 'Why not' (3.8), 'How to' (3.6), and 'What if' (4.1). Notably, the 'What if' explanation consistently received the highest ratings.

Supplementary Fig. 4 provides a visual summary of clinicians' mental model difference before and after explanations, along with their trust in the RF model following different types of SHAP-based explanations. Although each subplot presents the same data — normalized mental model difference (y-axis) and trust scores after explanations (x-axis) — they are separated by explanation type to illustrate differences in satisfaction, indicated by color intensity. This figure complements the quantitative results by demonstrating how explanation types relate to clinicians' understanding and trust. While mental model difference consistently decreased after explanations, trust scores varied across individuals. Some clinicians reported relatively low trust despite improved alignment, whereas others showed high trust even when mental model difference remained.

3.4. Summary of participant interviews

Clinicians provided qualitative feedback on the RF model explanations. The 'Why' explanation was perceived as straightforward but lacked detail regarding differing clinical viewpoints. The 'Why not' explanation was commended for clearly visualizing key features, although interpreting the color-coded data posed challenges for some clinicians. The 'How to' explanation was regarded as intuitively understandable, closely resembling EHR-derived metrics. The 'What if' explanation was considered the most persuasive, effectively integrating visual and numerical data, allowing clinicians to manipulate inputs and observe changes in prolonged ICU LOS risks.

However, presenting all 57 variables simultaneously was identified as overwhelming and time-consuming. Clinicians suggested real-time implementation of AI models could enhance clinical decision-making, understanding of patient conditions, and resource management. Recommendations for improvement included direct EHR integration for variable manipulation, comprehensive user training, and enhanced visual functionalities.

4. Discussion

This study highlights the pivotal role of XAI in healthcare by evaluating its impact on clinicians' acceptance of AI-generated decision-making. After reviewing the RF model explanations, clinicians' mental models became more aligned with the RF model. Initially, clinicians

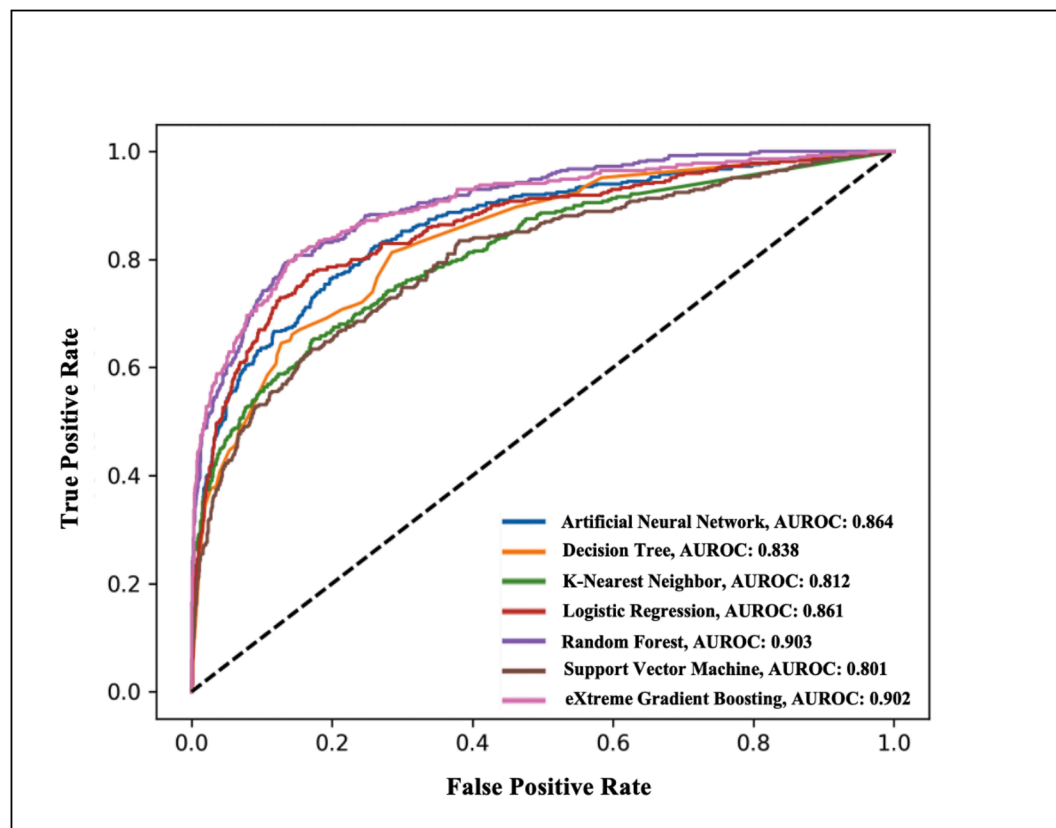


Fig. 1. Model performances by prediction models.

prioritized features related to ICU LOS based on their clinical experience. However, after explanations, there was a notable shift toward features identified by the RF model, suggesting that clinicians adopted a more data-driven approach, incorporating patient-specific characteristics alongside experience-based judgment. Despite this shift, clinicians' mental models did not fully align with the RF model, nor did they completely endorse AI-generated outcomes.

Our findings indicate that clinicians do not entirely abandon their clinical experience and intuition when incorporating AI models. Clinicians integrate AI recommendations while continuing to value their expertise, highlighting the importance of complementing rather than replacing clinical judgment with AI. Ongoing, multi-dimensional strategies are required to bridge the gap between AI models and clinician judgment. These strategies include developing functionalities that address user needs based on clinical context and conditions, and establishing feedback mechanisms, such as Human-In-The-Loop Machine Learning (HITL-ML), to support continuous user-system interaction [32–34]. Previous studies have demonstrated the effectiveness of HITL-ML in enhancing AI-driven medical data analysis by leveraging human expertise into the machine learning process [35–38]. This active interaction allows AI models to handle ambiguous cases more effectively by incorporating real-time clinician feedback, thus increasing model robustness and context-awareness [34].

Furthermore, developing a Human-Machine Collaborative Decision-making (HMCD) system that seamlessly integrates artificial and human intelligence is imperative to achieve optimal outcomes, and fosters a productive and complementary relationship between the AI model and clinicians [39]. Recent studies showed that the role of AI in healthcare is evolving, with growing recognition that human-AI collaboration is emerging as a promising approach [40–42]. This tailored approach has been proven to significantly enhance medical decision-making, highlighting the critical importance of integrating human expertise with AI to improve clinical outcomes and advance healthcare practices [43].

Trust in the RF model increased significantly after receiving the four SHAP-based explanations, with an average improvement of 1.1 points per item on a 5-point Likert scale, indicating that explanations enhance trust more effectively than presenting results alone. As a prerequisite for clinicians to better accept and integrate the AI model into their practice, the increase in trust is crucial for the effective implementation of the AI model. Additionally, clinicians' satisfaction with model explanations varied, with the 'What if' explanation receiving the highest satisfaction rating, averaging 4.1 points on a 5-point Likert scale. This high satisfaction suggests that interactive explanations allowing clinicians to explore potential scenarios are particularly beneficial. The 'How to' explanation was rated slightly lower in usefulness than the 'Why' explanation by 0.1 points, suggesting clinicians found identifying important features more helpful. However, the 'How to' explanation scored 0.1 points higher in comprehensiveness than the 'Why not' explanation, possibly due to the complexity of interpreting feature direction and color coding.

These findings suggest that trust in the AI model depends not only on cognitive alignment, but also on clinicians' satisfaction with the explanations and how they are delivered (Supplementary Fig. 4). Improving the content, depth, and format of explanations may therefore enhance both trust and satisfaction, ultimately supporting greater clinical acceptance of AI models. In particular, when clinicians encounter predictions that differ from their expectations, explanations play a key role in helping them evaluate the model's reasoning. Amann et al. (2020) demonstrated that when an AI model's recommendations significantly deviate from a clinician's expectations, explainability helps to verify the AI's parameters are clinically appropriate [44]. Explanatory methods that engage clinicians interactively, enabling effective consideration of future scenarios and deeper understanding of feature impacts, are essential [45].

This study has several limitations. First, the ICU LOS prediction model was based on data from a single hospital's surgical ICU, limiting

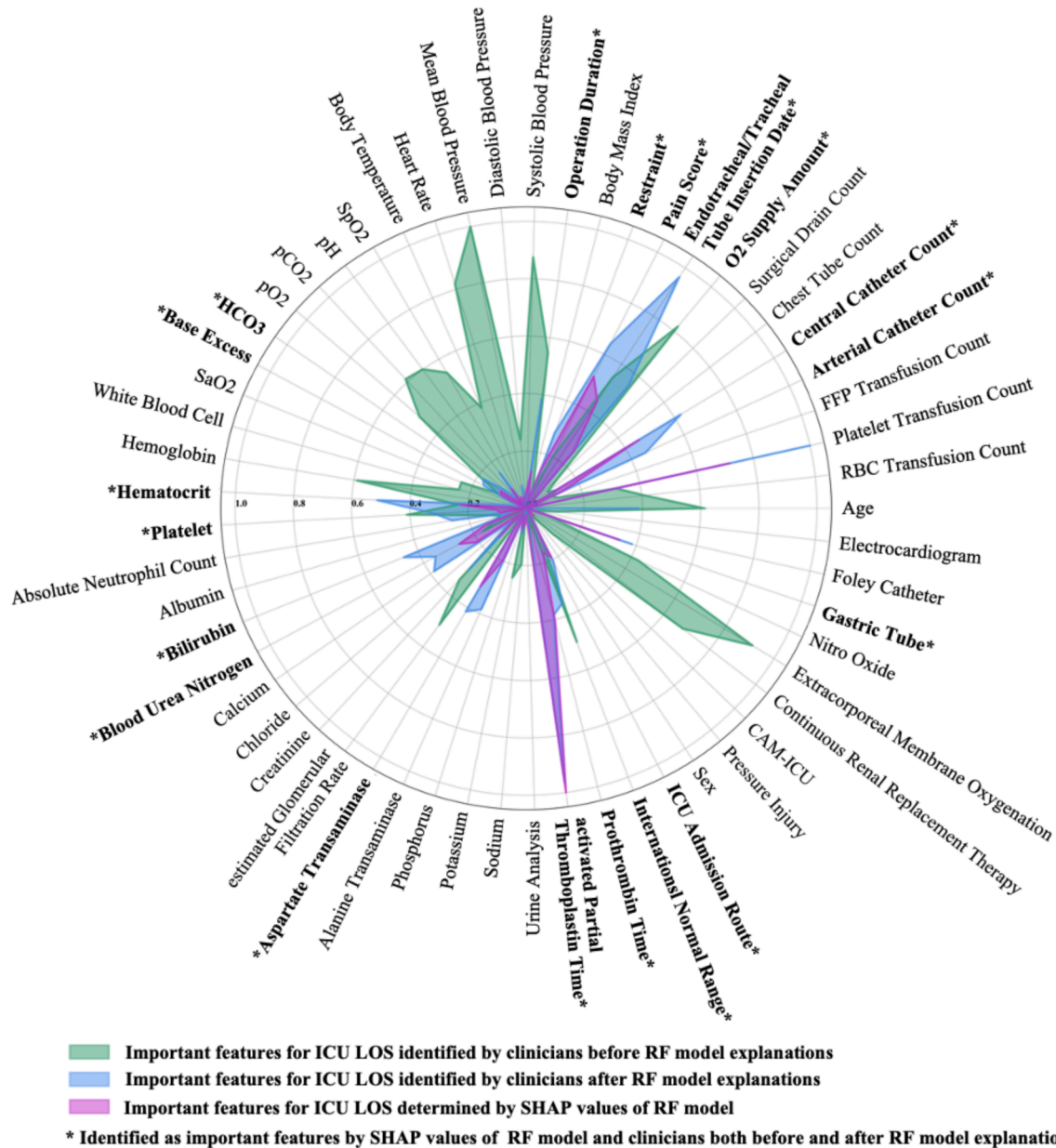


Fig. 2. Important features identified by the RF model and clinicians.

the generalizability and applicability of the results to other hospitals or ICU settings. Validation with data from diverse ICUs is needed to improve reliability. Second, a radar chart was used to compare important features between the AI model and clinicians, but this may not fully capture the complexity of clinicians' mental models. Additionally, normalizing important features and Shapley values to a 0–1 scale requires careful interpretation due to differing criteria. Developing more refined methods to assess mental models in clinical decision-making is necessary. Third, the small sample size of fifteen clinicians, with a higher ratio of nurses to physicians, may have introduced bias, limiting explanation effectiveness across specialties. A larger, more balanced sample of clinicians is recommended. Lastly, although binary classification showed strong predictive performance, categorizing ICU LOS into

multiple LOS intervals may provide more nuanced insights into the variations of LOS and more detailed predictive capabilities. Future research incorporating multiple LOS categories could provide greater clinical relevance and facilitate more tailored patient management strategies.

5. Conclusion

This study investigated the impact of XAI on clinicians' acceptance of AI-generated results by analyzing changes in their clinical decision-making processes (i.e., "mental models"), trust in the AI model, and satisfaction with explanations based on the DARPA's XAI evaluation framework. To further advance XAI integration in clinical practice,

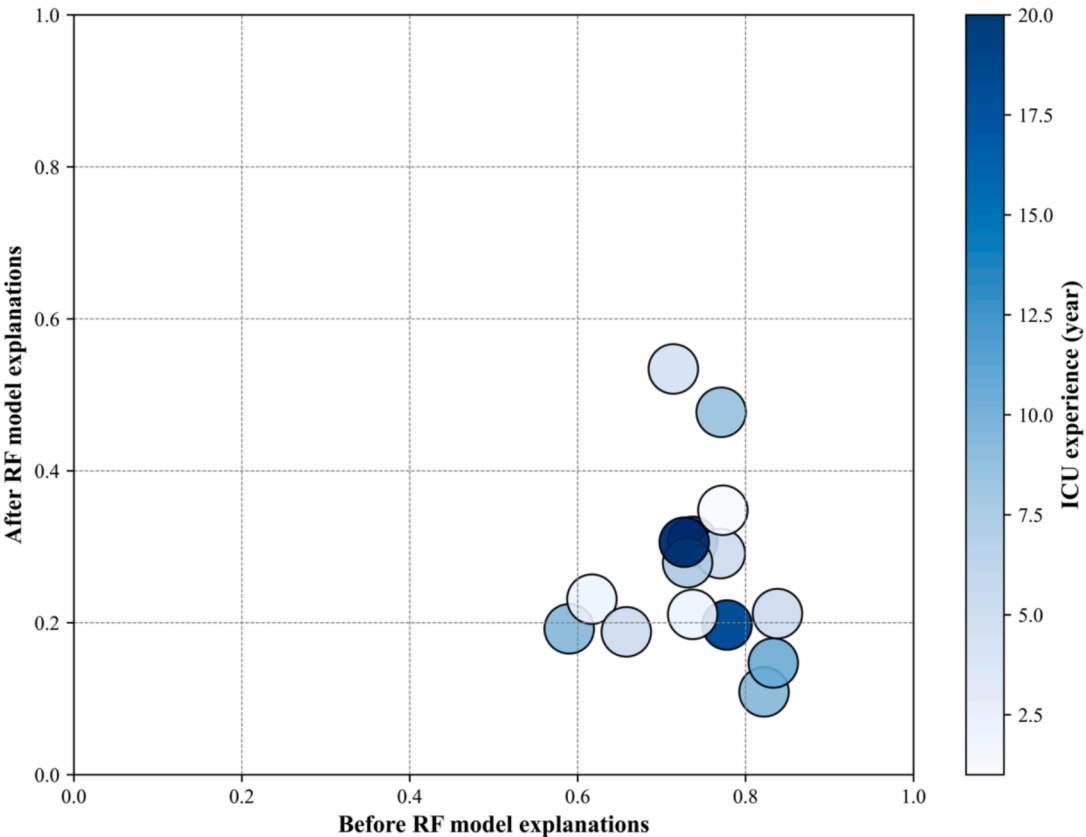


Fig. 3. Normalized difference in the feature importance rankings between the RF model and clinicians with varying ICU experience, before and after model explanations (higher = greater difference).

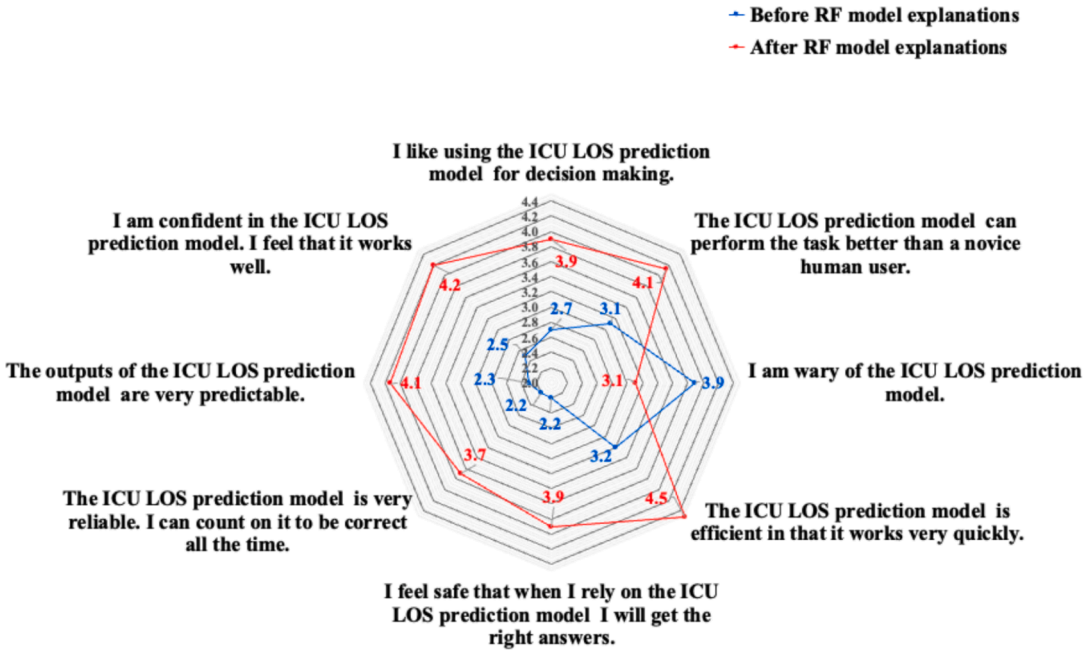


Fig. 4. Clinicians' trust in the RF model before and after model explanations.

continued efforts should focus on developing more sophisticated methods for assessing mental models in clinical decision-making. Additionally, ongoing evaluation of XAI impact on clinician acceptance of AI results through larger-scale studies is essential to establish a robust foundation for its effective implementation in real-world settings.

6. Code availability

The Python and R Shiny codes used in this study are available at <https://github.com/nursejung/Evaluating-the-Impact-of-Explainable-AI-on-Clinicians-Decision-Making>.

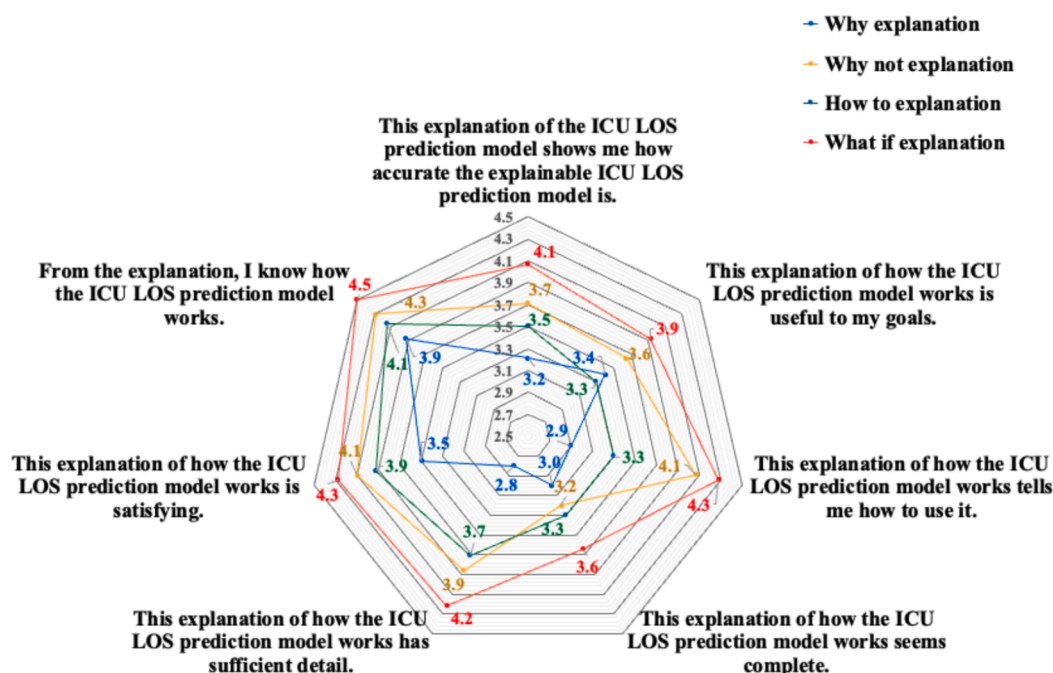


Fig. 5. Clinicians' satisfaction with four different model explanations.

Ethical statement

This study was approved by the Research Ethics Committee of Seoul National University Hospital (Ethics Review and Approval Number H-2212-053-1384) and Seoul National University (Ethics Review and Approval Number 2310/004-014), Seoul, South Korea. Informed consent was obtained from all participants during the recruitment phase.

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CRedit authorship contribution statement

Jinsun Jung: Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Sunghoon Kang:** Writing – review & editing, Visualization, Software, Investigation. **Jeeyae Choi:** Writing – review & editing, Supervision. **Robert El-Kareh:** Writing – review & editing, Supervision. **Hyungbok Lee:** Resources, Writing – review & editing. **Hyoneui Kim:** Writing – review & editing, Supervision, Conceptualization.

Data availability

The datasets underlying this article are available from the corresponding author upon reasonable request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ijmedinf.2025.105943>.

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